

Pandas DataFrame, Handling Missing Values & Built-in Functions



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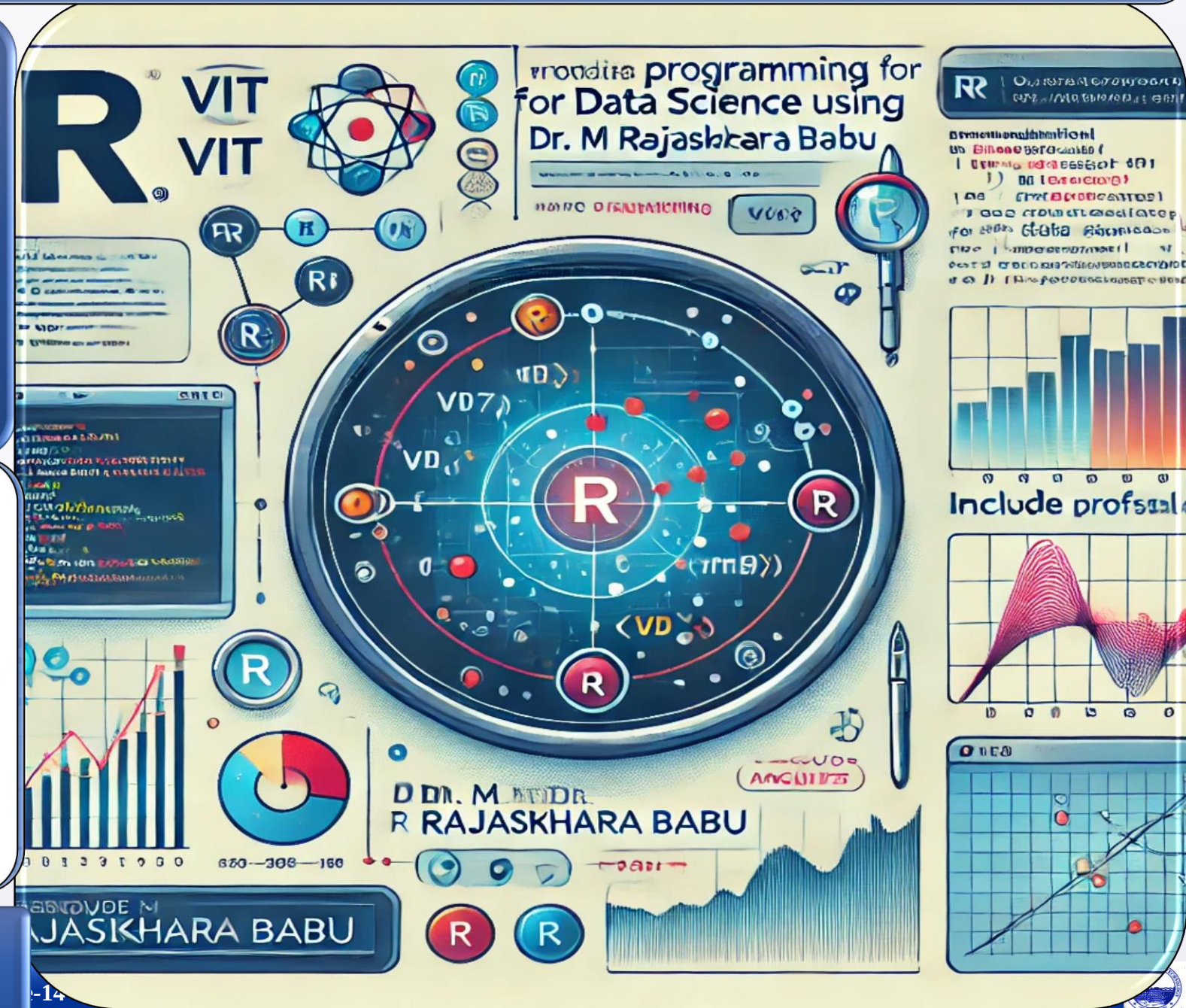


VIT[®]

Vellore Institute of Technology

(Deemed to be University under section 3 of UGC Act, 1956)

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Objectives & Teaching Learning Material



To explain the structure, creation, and manipulation of Pandas DataFrames for managing tabular data.



To demonstrate techniques for detecting, filling, and removing missing values in datasets using Pandas functions.



To illustrate the use of built-in Pandas functions for exploring, summarizing, and analyzing data efficiently.

Teaching + Learning

Material:

LCD, White board Marker, Presentation slides

Session Plan

Time in minutes	Topic	Learning Aid & Methodology	Faculty Approach	Student Activity	Skill and Competency Developed
05	Re-Cap	Quiz Presentation	Questions Organizes	Answers Identifies	Knowledge Understanding
10	Pandas DataFrame Fundamentals	Presentation	Presentation	Listens	Remembering
10	Creating, Accessing, and Understanding DataFrames	Explains	Explains	Notes	Understanding
15	Handling Missing Values in Pandas	Case Study	Organizes	Observes	Applying
10	Built-in Functions for Data Exploration and Analysis	Discussion	Facilitates	Answers	Analyzing

Definition

A DataFrame is a two-dimensional labeled data structure in Pandas — similar to a spreadsheet or database table.

— It is the most commonly used Pandas data structure.

- Rows and columns with labels
- Each column can hold different data types
- Supports powerful data manipulation

Basic Structure

Index	Name	Age	Marks
0	John	20	85
1	Mary	21	90
2	David	19	78

Import Pandas

```
import pandas as pd
```

Creating a DataFrame from a Dictionary

The most common way to create a DataFrame is from a Python dictionary, where keys become column names and values become column data.

```
#Load library
import pandas as pd
#Create dictionary for a table
data = {
    'Name': ['John', 'Mary', 'David'],
    'Age': [20, 21, 19],
    'Marks': [85, 90, 78]
}
#Convert Dictionary to dataframe using pandas
df = pd.DataFrame(data)
print(df)
```

Name	Age	Marks
John	20	85
Mary	21	90
David	19	78

Code

```
students = [  
    ['John', 20, 85],  
    ['Mary', 21, 90],  
    ['David', 19, 78]  
]  
  
df = pd.DataFrame(  
    students,  
    columns=['Name', 'Age', 'Marks']  
)  
print(df)
```

Name	Age	Marks
John	20	85
Mary	21	90
David	19	78

Advantages

Tabular Form

Organizes data in rows and columns for clarity

Easy Analysis

Enables fast filtering, grouping, and aggregation

Efficient Storage

Optimized memory use and fast manipulation

Accessing Data in a DataFrame

1

Single Column

```
df['Name']
```

Returns a Series with all Name values.

2

Multiple Columns

```
df[['Name','Marks']]
```

Returns a DataFrame with selected columns.

3

Row by Label

```
df.loc[0]
```

Access a row using its index label.

4

Row by Position

```
df.iloc[1]
```

Access a row by integer position — returns Mary's record.

Pandas DataFrames expose several built-in attributes that reveal the structure and metadata of your data — no function call needed.

Key Attributes

- `df.shape`
Returns (rows, columns)
- `df.size`
Total number of elements
- `df.columns`
Index of column names
- `df.index`
Row index labels
- `df.dtypes`
Data type of each column
- `df.ndim`
Number of dimensions (always 2)

Example Output

```
print(df.shape)
# (3, 3)

print(df.columns)
# Index(['Name', 'Age', 'Marks'])
```


Handling Missing Values

Missing values occur when data is unavailable or unknown. In Pandas, they are represented as NaN (Not a Number) or None.

```
import numpy as np
#Create a dictionary
data = {
    'Name': ['John', 'Mary', 'David'],
    'Marks': [85, np.nan, 78]
}

df = pd.DataFrame(data)
```

Index	Name	Marks
0	John	85.0
1	Mary	NaN
2	David	78.0

`isnull()` — Find Missing

```
df.isnull()
```

Returns a boolean DataFrame —
True where values are missing.

`notnull()` — Find Present

```
df.notnull()
```

Returns True where values are present.

Count Missing Per Column

```
df.isnull().sum()
```

Output

```
Name    0
Marks    1
dtype: int64
```

This tells us exactly which columns have missing data and how many gaps exist — essential before any cleaning step.

Filling Missing Values with `fillna()`

Replace with Zero

```
df.fillna(0)
```

Replaces all NaN values with 0.
Simple but may distort statistics.

Replace with Mean

```
df['Marks'].fillna(  
    df['Marks'].mean(),  
    inplace=True  
)
```

Fills NaN with the column average —
preserves statistical balance.

State	John	Mary	David
Before	85	NaN	78
After	85	81.5	78

Remove Rows with NaN

```
df.dropna()
```

Drops any row that contains at least one missing value.

Remove Columns with NaN

```
df.dropna(axis=1)
```

Drops any column that contains missing values.

Before vs. After

Name	Marks	Status
John	85	✓ Kept
Mary	NaN	✗ Dropped
David	78	✓ Kept

☐ Use `dropna()` when missing data is too sparse to impute reliably.

Built-in Functions Overview



Viewing Data

`df.head()` — first 5 rows

`df.tail()` — last 5 rows



Dataset Info

`df.info()` — schema, types, non-null counts, memory



Statistics

`df.describe()` — count, mean, std, min, max, quartiles



head()

`df.head()`

Displays the first 5 rows of the DataFrame.

Pass a number to customize:

`df.head(3)`.

tail()

`df.tail()`

Displays the last 5 rows. Useful for checking the end of large datasets.

Sample Output

Name	Age	Marks
John	20	85
Mary	21	90
David	19	78

These functions are the quickest way to sanity-check your data immediately after loading it.

The `df.info()` function prints a concise summary of the DataFrame's structure — invaluable for understanding what you're working with before analysis.



Row Count

Total number of entries in the index



Column Count

Number of columns and their names



Data Types

dtype of each column (int, float, object)



Non-Null Counts

How many non-missing values per column



Memory Usage

Total memory consumed by the DataFrame

```
df.info()
```

```
# RangeIndex: 3 entries
```

```
# Data columns (total 3 columns)
```

Usage

```
df.describe()
```

Generates descriptive statistics for all numeric columns automatically.
Provides a quick statistical snapshot of your data.

Metrics Provided

01 **Count**

Non-null value count

02 **Mean**

Average value

03 **Std**

Standard deviation

04 **Min / Max**

Extreme values

05 **Quartiles**

25%, 50%, 75% percentiles

Example Output (Marks)

Statistic	Value
count	3
mean	84.33
std	6.03
min	78
max	90

Additional Useful Functions

`df.columns`

Returns the list of all column names in the DataFrame.

`df.dtypes`

Shows the data type of each column — int64, float64, object, etc.

`df.shape`

Returns a tuple (rows, columns) representing the DataFrame's dimensions.

`df.count()`

Counts non-null values in each column — useful for spotting gaps.

`df.nunique()`

Returns the number of unique values per column — great for categorical analysis.

Practice Exercise



Build and explore a Student Performance Dataset with intentional missing values.

```
import pandas as pd, numpy as np
```

```
data = {  
    'Name': ['John','Mary','David','Lisa'],  
    'Marks': [85, np.nan, 78, 92],  
    'Age': [20, 21, 19, np.nan]  
}  
df = pd.DataFrame(data)
```

1 Display the DataFrame

Print the full DataFrame to inspect its structure.

2 Identify & Count Missing Values

Use `isnull()` and `isnull().sum()` to locate and count gaps.

3 Fill Missing Marks

Replace NaN in Marks using the column mean with `fillna()`.

4 Remove Remaining NaN Rows

Apply `dropna()` to remove rows still containing missing values.

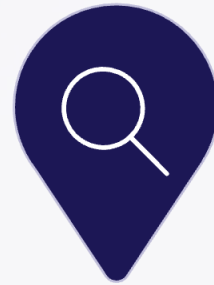
5 Explore with Built-in Functions

Use `head()`, `tail()`, `describe()`, and `info()` to summarize the cleaned data.

Missing Values — Full Workflow

Detect

Find nulls with
isnull, notnull, sum



Remove

Drop missing rows or
columns with dropna



Handle

Impute with fillna
using zero or mean

This three-step workflow — detect, handle, remove — is the standard approach to cleaning missing data in any Pandas project before analysis or modeling begins.

DataFrame Basics

- Created from dict or lists
- Accessed via `loc` / `iloc`
- Key attributes: `shape`, `dtypes`, `columns`

Missing Values

- Detected with `isnull()`
- Filled with `fillna()`
- Removed with `dropna()`

Built-in Functions

- `head()` / `tail()`
- `info()`
- `describe()`
- `count()` / `nunique()`



- A DataFrame is the core Pandas data structure for storing and analyzing tabular data.
- Built-in functions and missing-value handling tools help you clean and explore datasets efficiently — the essential first step in any data science workflow.

Create

Build DataFrames from dicts or lists

Clean

Detect and handle missing values

Explore

Summarize with built-in functions

- **Introduction to Pandas DataFrame**
 - Definition and characteristics of a DataFrame
 - Structure of rows, columns, and labels
 - Importing Pandas and basic concepts
- **Creating and Accessing DataFrames**
 - Creating DataFrames from dictionaries
 - Creating DataFrames from lists
 - Accessing columns and rows using indexing, loc, and iloc
- **DataFrame Attributes and Metadata**
 - shape, size, columns
 - index, dtypes, ndim
 - Understanding DataFrame structure and properties
- **Handling Missing Values**
 - Introduction to missing values (NaN, None)
 - Detecting missing values using isnull() and notnull()
 - Filling missing values using fillna()
 - Removing missing values using dropna()
- **Built-in Data Exploration Functions**
 - head() and tail()
 - info()
 - describe()
 - count() and nunique()
- **Data Cleaning Workflow and Practice Exercises**
 - Detecting, handling, and removing missing values
 - Exploring cleaned datasets using built-in functions
 - Student performance dataset case study

Reference

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Bharti Motwani, Data Analytics using Python – By Wiley (2020)

Jules S. Damji, Learning Spark: Lightning-Fast Data Analytics, Second Edition – By Shroff/O'Reilly (2020)

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